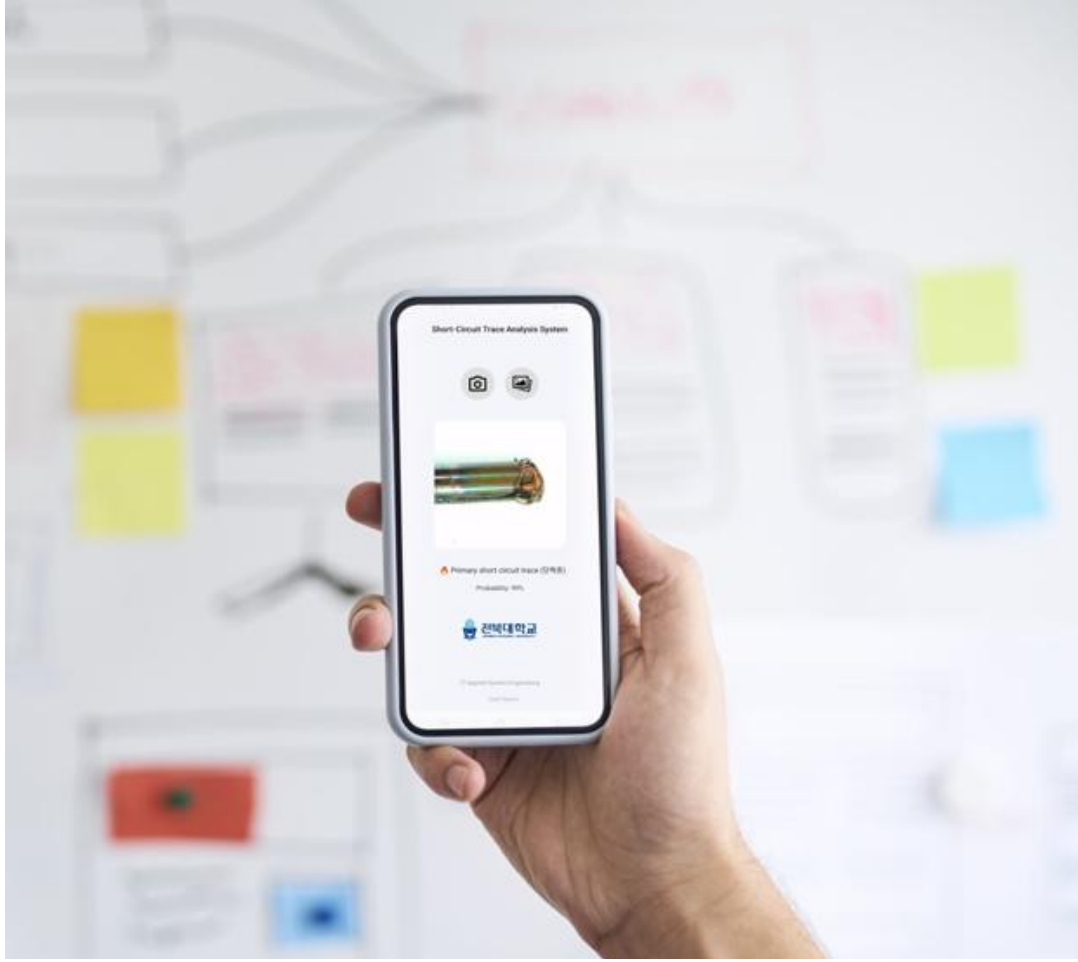
MOBILE DEEP LEARNING INFERENCE FRAMEWORK FOR ANDROID-BASED SHORT-CIRCUIT TRACE CLASSIFICATION



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JEONBUK NATIONAL UNIVERSITY

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INTRODUCTION

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Electrical fires often caused by short circuits

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- Need to distinguish PSCT vs SSCT

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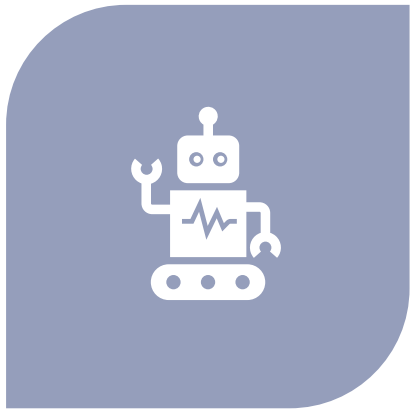
- Traditional methods are manual and subjective

04

- AI enables automated classification

OBJECTIVE



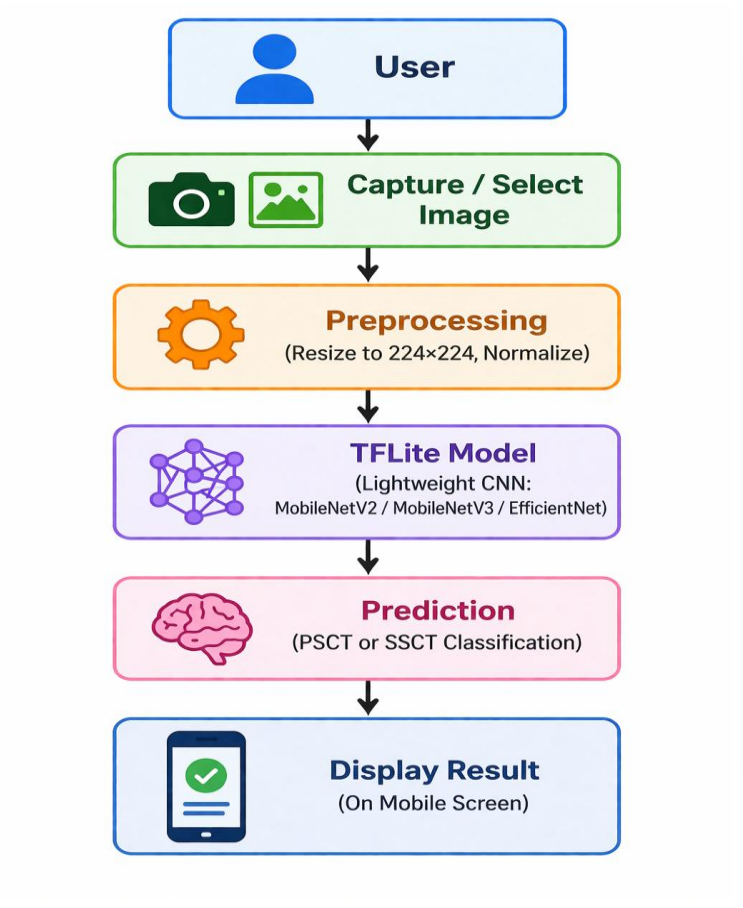
• DEVELOP ANDROID-BASED CLASSIFICATION SYSTEM

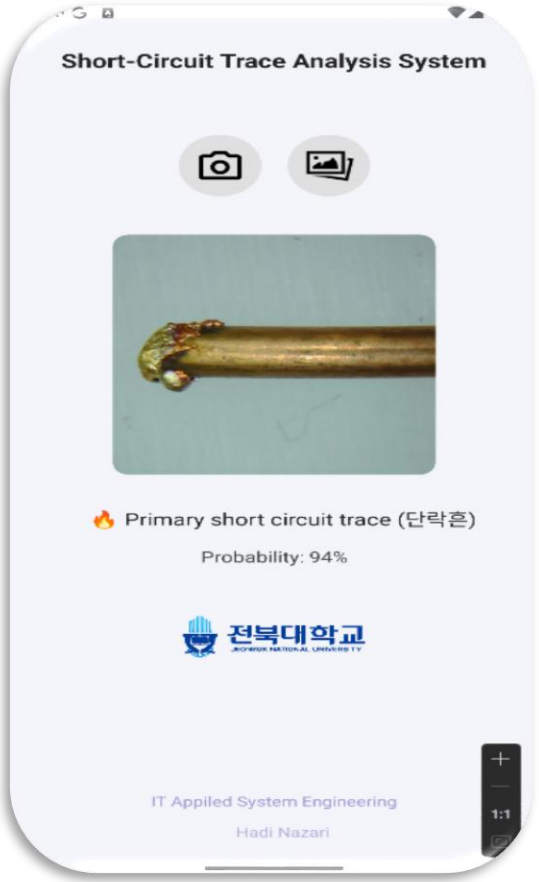
• USE LIGHTWEIGHT CNN MODELS

• ENABLE REAL-TIME MOBILE INFERENCE

PROPOSED FRAMEWORK FOR ELECTRIC FIRE SHORT-CIRCUIT TRACE CLASSIFICATION

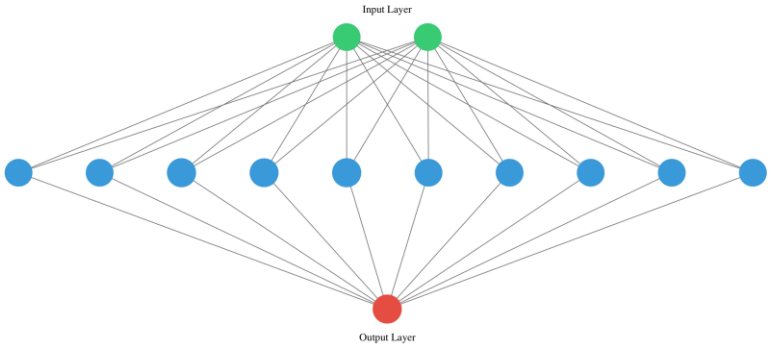
ANDROID SYSTEM OVERVIEW





ANDROID IMPLEMENTATION

- The mobile application was developed using:
  - **Java**
  - **Android Studio**
- Model trained in Python (Anaconda, TensorFlow)



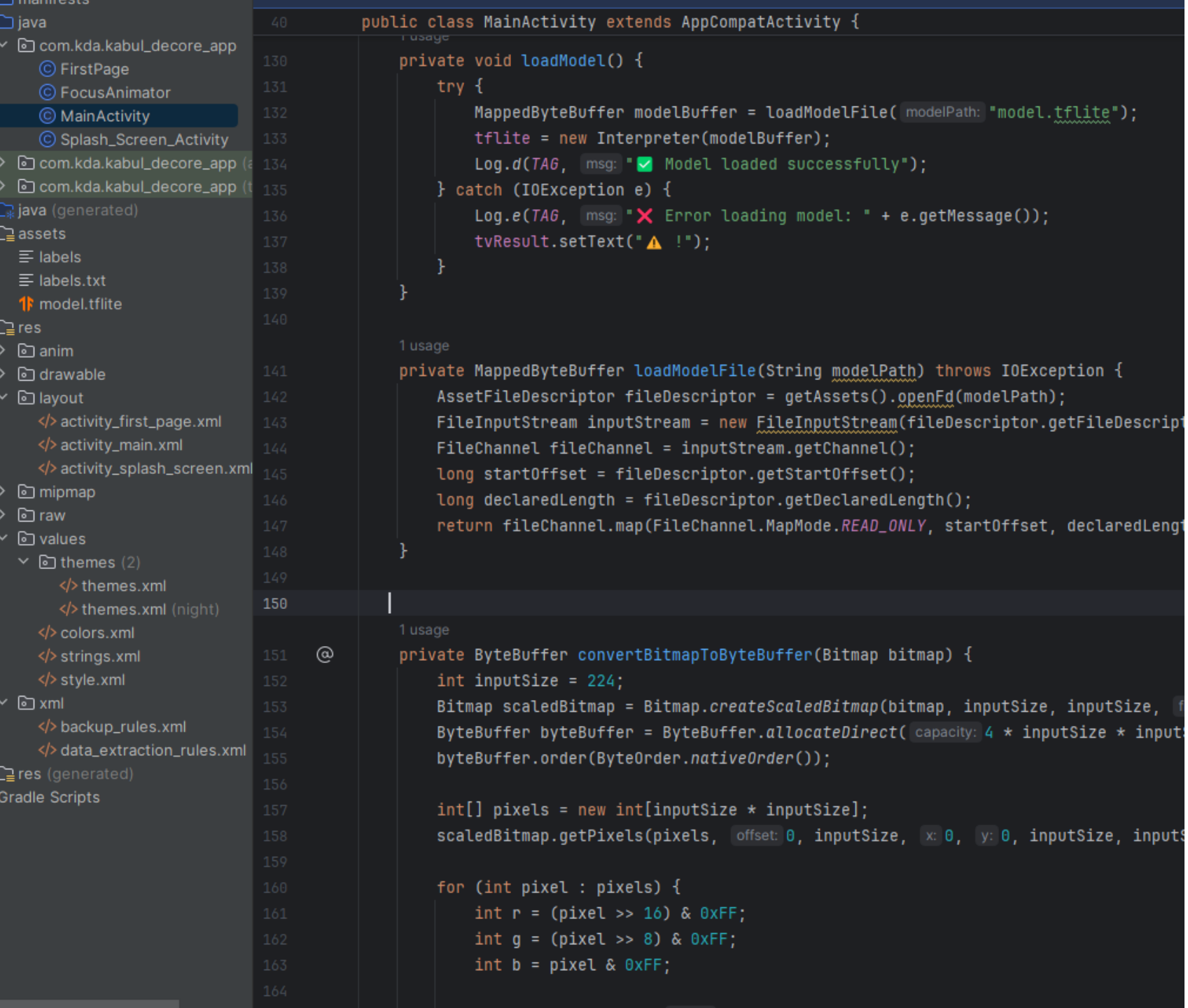
DATASET

Total images: 752

• Classes: primary short-circuit traces (PSCT), secondary short-circuit traces (SSCT)

• Split: 80% training, 20% testing

• Augmentation: rotation, zoom, flip, brightness



PREPROCESSING

- Input image resized to: 224 × 224 pixels
- Converted into ByteBuffer format
- Pixel normalization applied for model compatibility

TensorFlow

MODEL CONVERSION (TENSORFLOW LITE)

• After training, the best-performing model was converted to TensorFlow Lite (.tflite) format.

• This step enables:

- •Reduced model size
- •Faster inference
- •Compatibility with mobile devices

INFERENCE PROCESS

Input image converted into numerical tensor

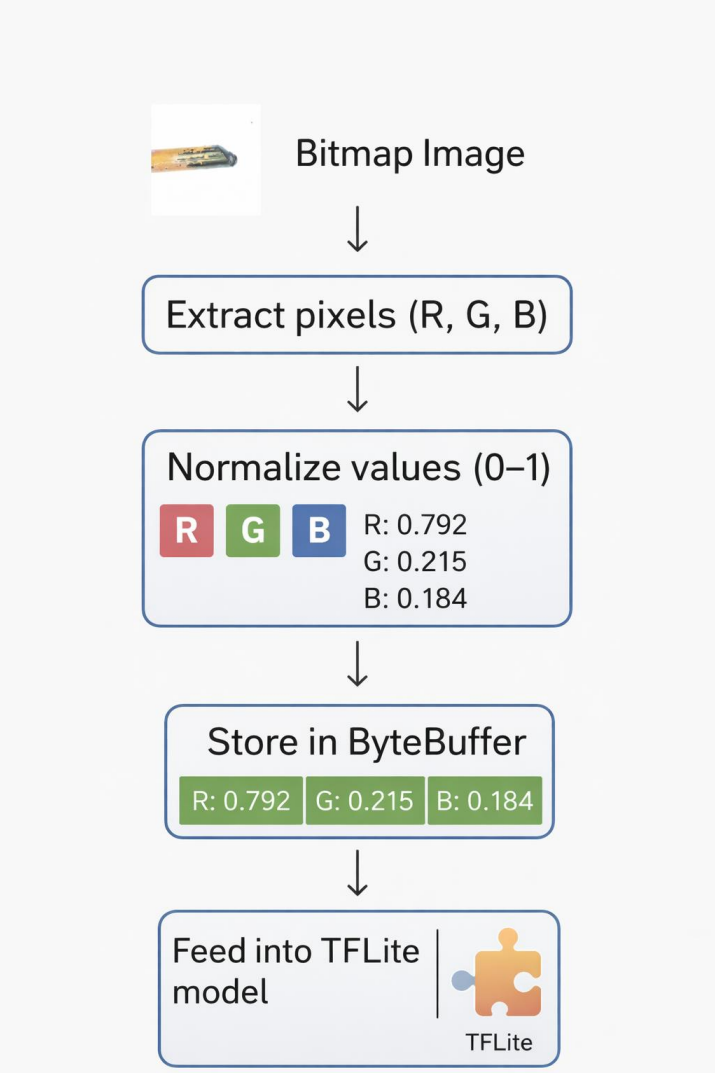
Model outputs a probability value:

$< 0.5 \rightarrow$ PSCT

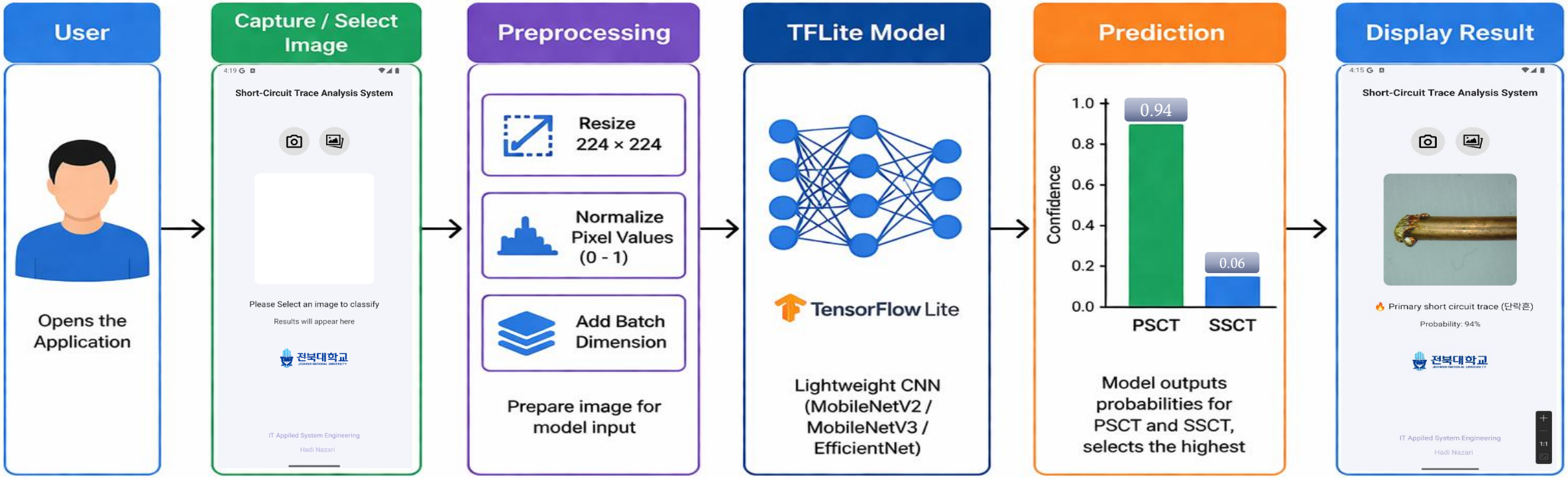
$\geq 0.5 \rightarrow$ SSCT

BYTEBUFFER

- A **memory container for raw numerical data**
- TensorFlow Lite understands only numerical arrays (tensors).
- TensorFlow Lite requires input in this form because:
  - It is **fast**
  - It is **low-level (close to hardware)**
  - It avoids extra memory copying



SYSTEM OVERVIEW

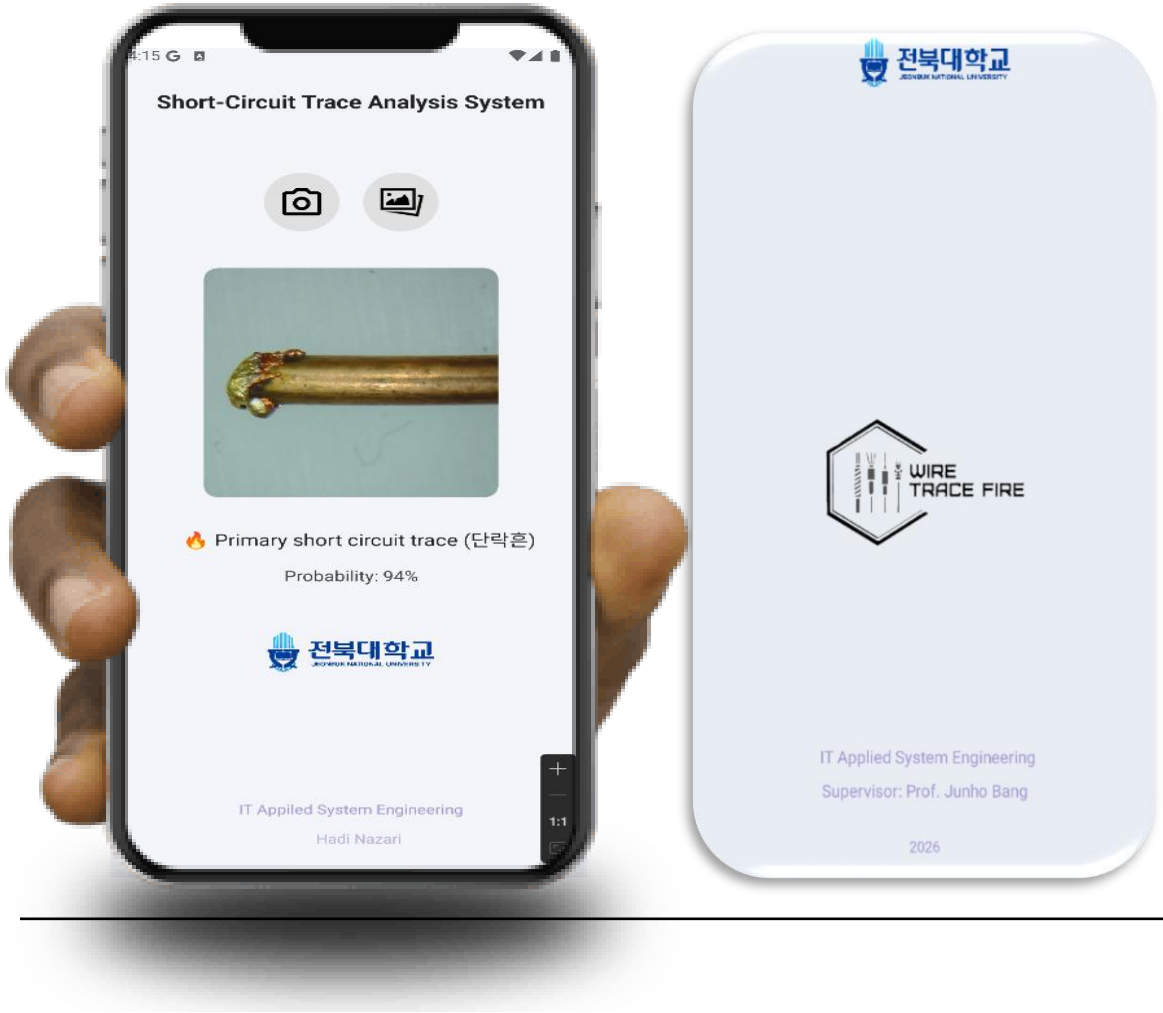


RESULTS

EfficientNet: 96.05% accuracy

- MobileNetV3: stable performance
- MobileNetV2: efficient and fast

APPLICATION INTERFACE



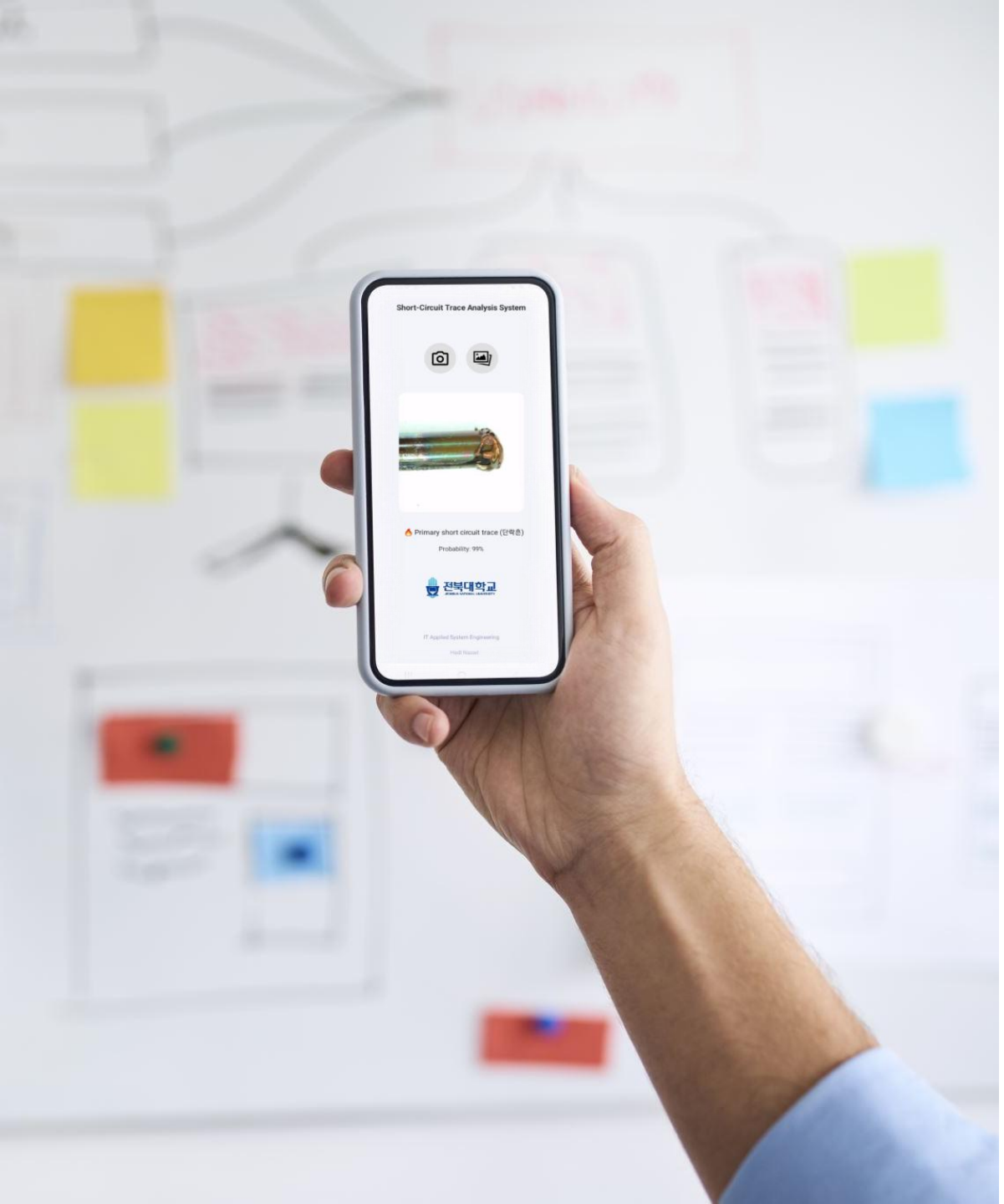
- Image input (camera/gallery)
- Prediction result
- Probability output
- User-friendly UI

ADVANTAGES OF THE MOBILE IMPLEMENTATION

- Real-time classification
- No internet required (on-device inference)
- Lightweight and efficient
- Suitable for field use (fire investigation)

SUMMARY

- Model trained in Python (Anaconda, TensorFlow/Keras)
- Converted to TensorFlow Lite format
- Integrated into Android Studio (Java)
- Supports camera & gallery input
- Performs real-time on-device classification
- Displays predicted class and confidence



FUTURE WORK

- ❖Expanding the dataset to improve generalization
- ❖Enhancing accuracy using advanced models
- ❖Improving the user interface with better graphics and usability
- ❖Adding more features for practical and flexible use

THANK YOU

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